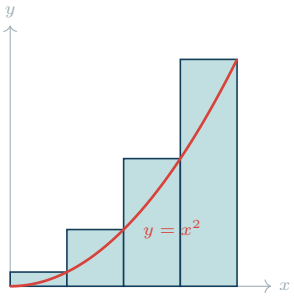
SEQUENCES LADDER 8 OF 9

From sums to the integral

Adding more and thinner rectangles turns the sums of powers into the area under a curve.

1 / THE RULE AND THE PICTURE

Thin rectangles under $y = x^2$ add up to $\frac{1}{3}$



four rectangles: $\frac{1}{4}\left(\frac{1}{1}6 + \frac{4}{1}6 + \frac{9}{1}6 + 1\right) = 0.469$

Split $0 \le x \le 1$ into n strips of width $\frac{1}{n}$ and take the height at the right end of each, so the area is $\frac{1}{n}\sum_{k=1}^{n}\left(\frac{k}{n}\right)^2 = \frac{n(n+1)(2n+1)}{6n^3}$, and as n grows

$$\frac{(n+1)(2n+1)}{6n^2} \to \frac{1}{3} = \int_0^1 x^2\,dx$$

The formula for the sum of squares from the earlier sheet does all the work.

2 / NUMBERS

The totals settle on $\frac{1}{3}$

rectangles n	total area	excess over $\frac{1}{3}$
4	0.46875	0.1354
10	0.385	0.0517
100	0.33835	0.0050
1000	0.3338335	0.0005
$\to \infty$	$\frac{1}{3}$	0

The excess over $\frac{1}{3}$ is $\frac{1}{2n} + \frac{1}{6n^2}$, so it shrinks like $\frac{1}{2n}$, and each tenfold increase in the number of rectangles gives one more correct digit.

3 / OTHER POWERS

The leading coefficient of the sum gives the area

For $y = x$ the sum is $\frac{n+1}{2n}$, which is 0.625 for $n = 4$ and tends to $\frac{1}{2}$, and for $y = x^3$ it is $\frac{(n+1)^2}{4n^2}$, which is 0.3906 for $n = 4$ and tends to $\frac{1}{4}$. In general $\frac{1}{n^{p+1}}\sum k^p \to \frac{1}{p+1}$ because the sum has leading term $\frac{n^{p+1}}{p+1}$, so $\int_0^1 x^p\,dx = \frac{1}{p+1}$.

4 / TELESCOPING BECOMES THE FUNDAMENTAL THEOREM

Adding differences gives the end value minus the start

Adding the first differences of any sequence collapses: $\sum_{k=a}^{b-1}\Delta f(k) = f(b) - f(a)$, as $\sum_{k=0}^{n-1}(2k+1) = n^2$ shows with $f = n^2$. The integral version is the same statement with differences replaced by derivatives, $\int_a^b f'(x)\,dx = f(b) - f(a)$, so finding an area means finding a function whose derivative is the height.

5 / THE AREA FUNCTION

Its gradient is the height

Let $A(x) = \int_0^x t^2\,dt = \frac{x^3}{3}$. Increasing x by a small h adds a thin strip of area about x^2h, so $\frac{A(x+h)-A(x)}{h} \to x^2$, which is $A'(x) = x^2$, and in the discrete case the totals of squares differ by exactly n^2 from one total to the next.

6 / TRY THESE

Work out the area

(a) Find the area under $y = x$ from 0 to 1 with 10 rectangles: $\frac{11}{20} = 0.55$, heading to $\frac{1}{2}$. **(b)** Check with the triangle: its area is $\frac{1}{2}\cdot 1\cdot 1$. **(c)** Using $\sum k^3 = \left(\frac{n(n+1)}{2}\right)^2$, show that the rectangles under x^3 give $\frac{(n+1)^2}{4n^2} \to \frac{1}{4}$.

ASIDE / ONE SIDED SUMS

Taking the height at the right end overestimates a rising curve and the left end underestimates it, and the true area sits between them.

LINKS TO OTHER SHEETS

The previous sheet gives the derivative from differences; the last sheet collects the whole dictionary.

Convention. n is the number of strips; the area is the limit of the rectangle sums; $\int_a^b$ is the integral from a to b.